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Path-following and LiDAR-based obstacle avoidance via NMPC for an autonomous surface vehicle

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ABSTRACT

This paper presents a path-following and collision avoidance system for autonomous surface vehicles based on nonlinear model predictive control. The proposed strategy is capable of following a desired path while maintaining a commanded velocity, whereas it can diverge from the references to safely perform maneuvers to elude unexpected obstacles. A nonlinear dynamic model of the vehicle is applied to predict the vehicle states into a finite horizon. Furthermore, a LiDAR sensor located at the front of the boat is employed for local object detection. Here, the high-performance optimal control framework, acados, is used to solve the optimization problem onboard. Real-time numerical simulations and field experiments demonstrate the effectiveness of the proposed approach against multiple buoys.

1. Introduction

Applications such as environmental monitoring, disaster management (Murphy et al., 2008), emergency response, offshore construction, inspection (Lindemuth et al., 2011), urban transportation (Wang et al., 2020), among others, could benefit from Autonomous Surface Vehicles (ASVs) (Murphy et al., 2008; Lindemuth et al., 2011; Wang et al., 2020; Liu et al., 2016). Nonetheless, these activities might involve complex, unstructured and congested environments. Motion planning and control systems should be capable of safely navigating through a variety of scenarios. In this sense, if a global path is known, path-following controllers can accurately drive the vehicle, independent of time constraints (Fossen, 2011). Then, collision avoidance systems or local planners, are relevant to counteract unexpected obstacles, which can deviate the desired trajectory and risk the safety of the vessel.

One of the commonly employed systems is the underactuated ASV (Karimi and Lu, 2021), which has less control inputs than state variables. Here, the path-following strategy involving the line-of-sight guidance law in cascade with a low-level controller, such as PID, is regularly adopted (Fossen, 2011). Nevertheless, the task is usually considered independent from local obstacle avoidance. Then, to solve collision avoidance, different strategies have been proposed. For instance, Velocity Obstacle (VO)-based approaches have been proposed (Campos et al., 2019; Huang et al., 2019; Kuwata et al., 2014). In Huang et al. (2019), a Generalized VO (GVO) considering slip dynamics is presented. This approach can be used to support close-range collision prevention with multiple obstacles, illustrated via a simulation case study of ship-encountering. However, the GVO required a cascaded scheme, adding a surge and yaw PD controller. Wiig et al. (2020) presented the Constant Avoidance Angle (CAA) algorithm, considering a singular obstacle shaped as a circle. The CAA switched with a nominal guidance law, such as Pure Pursuit or Line Of Sight (LOS), depending on the obstacle presence, and applied feedback linearizing controllers for surge and yaw. Experimental results with a full scale research ship validated the approach, and analyzed its implementation against multiple obstacles. Rapidly-exploring Random Tree was studied with numerical simulations in Chiang and Tapia (2018), computing a local trajectory for a controller to follow. In Wang et al. (2018b), a Local Normal Distribution-based Trajectory (LNDT) method was proposed, demonstrating its performance with numerical simulations. This approach then required a controller to follow the LNDT trajectory. A method based on region analysis for local reactive obstacle avoidance for high-speed Unmanned Surface Vehicles (USVs) was established in Tang et al. (2015). This guidance system has to be combined with a surge and yaw controller to complete the task. Simulations and sea-trials validated the performance against multiple

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obstacles, while the USV maneuvered at speeds higher than 20 knots. A waypoint-based strategy was experimentally implemented in Villa et al. (2020), creating local positional targets for a LOS-PID cascade path-following controller. Mousazadeh et al. (2018) used Potential Fields and the search ball algorithm for a catamaran USV obstacle avoidance. Here, a locally perceived obstacle and the path enter the avoidance algorithm, which in turn outputs the commands for two thrusters, making it an end-to-end avoidance and control solution. Experimental results validated the approach as the USV managed to avoid two obstacles located at different instances of a predefined path. Implementing Deep Reinforcement Learning is another strategy (Woo and Kim, 2020; Xu et al., 2020; Meyer et al., 2020). Meyer et al. (2020) presented an end-to-end approach, where the input of the neural network included the vehicle states, the desired path, and segmented sensor information (e.g. from a LiDAR), whereas the output consisted of the ASV surge force and yaw moment. Simulations following a path and avoiding multiple circle-shaped obstacles demonstrated the feasibility of the approach.

Furthermore, obstacle evasion and control can be addressed as an optimization problem (Collado-Gonzalez et al., 2021). In that sense, Model Predictive Control (MPC) and Nonlinear MPC (NMPC) methods optimize the control action in each time-step according to a cost function and are able to handle constraints related to the system dynamics or the environment (Sotelo et al., 2018, 2020). Hence, such control strategies have been applied to ASVs facing different constraints such as underactuation, actuator saturation or local obstacles. Nevertheless, MPC can be computationally expensive, which becomes a disadvantage in practical systems. To benefit from its characteristics, MPC was proposed in Guerreiro et al. (2014), Wang et al. (2018a, 2020) for trajectory tracking of fully-actuated ASVs, exhibiting both numerical simulations and real-time full scale experiments. Moreover, end-to-end trajectory tracking and collision avoidance for an underactuated ASV was studied in Abdelaal et al. (2018), using the ACADO toolkit for real-time implementation, but without field experiments.

Although trajectory tracking is useful for time-critical tasks, some activities require prioritizing the safety of the ASV against collisions, such as cluttered environments with tight spaces to maneuver. Then, the ASV should be allowed to operate without time constraints, or to temporally deviate from the original path if necessary. Thus, real-time path-following and local planning strategies are relevant. An MPC-based guidance law was introduced in Collado-Gonzalez et al. (2021) in combination with a robust controller for path-following of underactuated USVs, achieving real-time simulation results. However, collision avoidance and experimental validation were not presented. In Kufoalor et al. (2020), Hagen et al. (2018), Kufoalor et al. (2019), an MPC-based collision avoidance guidance system for underactuated vessels was presented with full scale experiments. Nonetheless, to reduce the computational burden, they only used a kinematic model, and a finite set of heading and surge speed references, instead of the full ASV dynamics and continuous actions, respectively. Likewise, an Automatic Identification System was used for obstacle detection in Kufoalor et al. (2020), Hagen et al. (2018), in contrast to the faster and more accurate radar-based solution from Kufoalor et al. (2019).

Therefore, the contribution of this paper is the proposal of a single algorithm solving collision avoidance, and path following control for ASVs based on NMPC. To the authors best knowledge, this is the first NMPC for covering end-to-end ASV obstacle avoidance and path-following. The advantage is that only one module is necessary, instead of the standardized three independent subsystems (collision avoidance, guidance law, and low-level controller), hence simplifying implementation and tuning. Additionally, continuous control of the surge speed allows to maintain a reference velocity, or decelerate/accelerate to enhance maneuverability. The high-performance embedded optimal control framework, acados (Verschueren et al., 2018), is employed to efficiently solve evasion, guidance, and control as an optimization problem online. A LiDAR-based obstacle detection system is additionally incorporated onboard. The proposed strategy is formulated on the nonlinear ASV dynamic model, a continuous set of actions, and a smooth thruster commands. Finally, simulations and real-time experiments demonstrate the advantages and effectiveness of the proposed method.

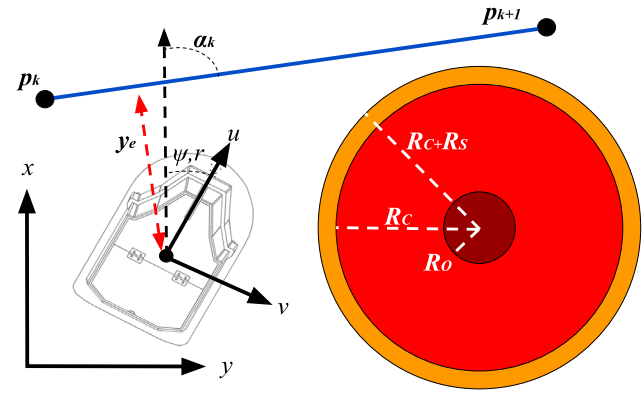

Fig. 1. Reference frames, path-following problem, and obstacle representation.

This paper is structured as follows: Section 2 addresses the problem formulation, whereas Section 3 presents the proposed path-following and collision avoidance strategy. Section 4 describes the practical implementation on the ASV platform, and experimental results are exhibited and discussed in Section 6. Finally, conclusions are given in Section 7.

2. Problem formulation

In this section, the problem (Fig. 1) is described. Suppose a mission planner or a global planner provides the underactuated ASV with waypoints and a reference velocity to achieve a task. Then, the ASV should safely navigate through the path created by such waypoints. Hence, a path-following and collision avoidance strategy is required. Path-following is defined as following a predefined path independent of time constraints (Fossen, 2011). In order to address the aforementioned, a new reference frame is created at the desired waypoint location $\boldsymbol{p}_k = [x_k, y_k]$, and the path angle α_k is introduced relative to the inertial frame. Now, the objective of the path-following controller is to minimize the distance between the ASV and the path, *i.e.*, to reduce the cross-tracking error

$$y_e = -[x - x_k]\sin a_k + [y - y_k]\cos a_k \tag{1}$$

where x and y represent the ASV position following the North-East-Down (NED) convention. To consider the kinematic effect from external disturbances such as currents, the course angle $\chi = \psi + \beta$ is defined, where ψ is the ASV yaw angle, $\beta = \text{asin}(v/U)$ is the sideslip angle, $U = \sqrt{u^2 + v^2}$, and u and v are the surge and sway velocities, respectively, in the body frame. Finally, in a real scenario, the ASV may encounter multiple obstacles which risk the safety of the vessel and the success of the task. Thus, it may be necessary to execute maneuvers and diverge from the desired path to avoid such obstacles.

3. NMPC-based path-following and collision avoidance

In this section, the considered mathematical model and the proposed NMPC-based path-following and collision avoidance method are presented.

3.1. Nonlinear state space model

The NMPC main control objective for path-following is to minimize the cross-tracking error y_e (1). Moreover, to ensure the ASV follows the path in the correct direction, it is also desired that the course angle follows the path angle ($\chi = \alpha_k$). Now, because the angles are wrapped $(-\pi, \pi]$, the NMPC has the goal of reaching $\sin\chi = \sin\alpha_k$ and $\cos\chi = \cos\alpha_k$, which respects the smallest-signed angle property. Here, since path-following is independent of time, the proposed strategy has the additional objective of maintaining a constant desired speed u_d. Thus, the equations which relate the problem and the ASV dynamics are

$$\dot{\chi} = r \tag{2}$$

$$\dot{\chi}_s = r\cos\chi \tag{3}$$

$$\dot{\chi}_c = -r\sin\chi \tag{4}$$

$$\dot{y}_e = -(u\cos\psi - v\sin\psi)\sin\alpha_k + (u\sin\psi + v\cos\psi)\cos\alpha_k \tag{5}$$

$$\dot{x} = u\cos\psi - v\sin\psi \tag{6}$$

$$\dot{y} = u\sin\psi + v\cos\psi \tag{7}$$

$$\dot{u} = \frac{1}{m - X_{\dot{u}}}\left[(m - Y_{\dot{v}})vr + X_{u|u|}u|u| + X_u u + \tau_u\right] \tag{8}$$

$$\dot{v} = X(u)r + Y(u)v \tag{9}$$

$$\dot{r} = \frac{1}{I_z - N_{\dot{r}}}\left[(Y_{\dot{v}} - X_{\dot{u}})uv + N_{r|r|}r|r| + N_r r + \tau_r\right] \tag{10}$$

$$X(u)r = \frac{1}{m - Y_{\dot{v}}}\left[(-m + X_{\dot{u}})ur\right] \tag{11}$$

$$Y(u)v = \frac{1}{m - Y_{\dot{v}}}\left[Y_{v|v|}v|v| + Y_v v\right] \tag{12}$$

where $\chi_s = \sin\chi$, $\chi_c = \cos\chi$, r is the yaw angular rate, Eqs. (2)–(5) represent the path-following error model, whereas Eqs. (6)–(10) include the underactuated ASV dynamic model (Gonzalez-Garcia and Castañeda, 2021). τ_u and τ_r are forces and moments given by

$$\tau_u = T_{\text{port}} + \varpi T_{\text{stbd}} \tag{13}$$

$$\tau_r = (T_{\text{port}} - \varpi T_{\text{stbd}})B/2 \tag{14}$$

where T_{port} and T_{stbd} are two thrusters located at the rear of the ASV, and $\varpi = 0.78$ is a correction factor required by the VTec S-III (Gonzalez-Garcia and Castañeda, 2021), which is the selected ASV for this work. The rest of the parameters relate to the vessel mass, moment of inertia, and hydrodynamics, with their identified values are shown in Tables 1 and 2 (Gonzalez-Garcia and Castañeda, 2021).

Then, the model can be rewritten into the nonlinear state space form:

$$\dot{\boldsymbol{x}} = f(\boldsymbol{x}, \boldsymbol{u}) \tag{15}$$

with $\boldsymbol{x} = [\chi, \chi_s, \chi_c, y_e, x, y, u, v, r]^T$ being the state, and $\boldsymbol{u} = [T_{\text{port}}, T_{\text{stbd}}]^T$ the control input. Nevertheless, with the aim to obtain smooth control signals, the state is augmented as follows $\boldsymbol{x} = [\chi, \chi_s, \chi_c, y_e, x, y, u, v, r, T_{\text{port}}, T_{\text{stbd}}]^T$, and $\boldsymbol{u} = [\dot{T}_{\text{port}}, \dot{T}_{\text{stbd}}]^T$, respectively. Thus, while avoiding obstacles, the main optimization objectives are:

$$\lim_{t\to\infty} y_e = 0 \tag{16}$$

$$\lim_{t\to\infty}(\sin\chi - \sin\alpha_k) = 0 \tag{17}$$

$$\lim_{t\to\infty}(\cos\chi - \cos\alpha_k) = 0 \tag{18}$$

$$\lim_{t\to\infty}(u - u_d) = 0 \tag{19}$$

Table 1
VTec S-III physical parameters.

Parameter	Value
Length (L)	1.01 [m]
Draft (T)	0.09 [m]
Centerline-to-centerline side hull separation (B)	0.41 [m]
Individual hull beam (B_{hull})	0.27 [m]
Mass (m)	30 [kg]
Moment of inertia (I_z)	4.1 [kg m^2]

Table 2
VTec S-III hydrodynamic parameters.

Coefficient	Correction factor	Dimensional term/equation
X_u		−25, ($u > 1.2$) 64.55
Y_v	0.5	$y(v)$
N_r	0.02	$-\pi\rho\sqrt{(u^2+v^2)}T^2L^2$
$X_{u\|u\|}$		0, ($u > 1.2$) −70.92
$Y_{v\|v\|}$		−99.99
$N_{r\|r\|}$		−3.49
$X_{\dot{u}}$		−2.25
$Y_{\dot{v}}$		−23.13
$N_{\dot{r}}$		−2.79

$$y(v) = -40\rho|v|\left[1.1 + 0.0045\frac{L}{T} - 0.1\frac{B_{hull}}{T} + 0.016\left(\frac{B_{hull}}{T}\right)^2\right]\left(\frac{\pi T L}{2}\right)$$

3.2. Nonlinear model predictive control

In the proposed approach, obstacles are added into the optimization as hard and soft constraints, where static obstacles are assumed to be circular, or fit into a circular area, represented in Fig. 1. Thus, to avoid obstacles, the following inequality must hold

$$\sqrt{(x_O^j - x)^2 + (y_O^j - y)} \geq R_C^j \tag{20}$$

where x_O^j and y_O^j represent the position of each obstacle j, and R_C^j is the collision radius of each obstacle defined as

$$R_C^j = R_O^j + R_V \tag{21}$$

Here, R_O^j stands for each obstacle radius, and R_V for the ASV radius. However, this hard constraint may risk collision of the ASV, as the NMPC will attempt to be as close to the obstacle as possible. Therefore, an arbitrary distance $R_S \in \mathbb{R}^+$ is introduced as a slack variable to increase the safety of the ASV. Hence, to achieve path-following and collision avoidance, the following constrained nonlinear program (NLP) must be solved

$$\begin{aligned} \min_{\substack{\boldsymbol{x}_0,\dots,\boldsymbol{x}_N,\\ \boldsymbol{u}_0,\dots,\boldsymbol{u}_{N-1}\\ \boldsymbol{z}_0,\dots,\boldsymbol{z}_{N-1}}} & \sum_{i=0}^{N-1} \|\boldsymbol{x}_i - \boldsymbol{x}_{i,\text{ref}}\|_Q{}^2 + \dots \\ & \dots + \|\boldsymbol{u}_i\|_R{}^2 + \|\boldsymbol{z}_i\|_S{}^2 + \dots \\ & \dots \|\boldsymbol{x}_N - \boldsymbol{x}_{N,\text{ref}}\|_{Q_N}{}^2 \end{aligned} \tag{22}$$

$$\begin{aligned} \text{s.t. } & \dot{\boldsymbol{x}}_i = f(\boldsymbol{x}_i, \boldsymbol{u}_i), && i = 0, \dots, N-1 \\ & u_{\min} \leq u \leq u_{\max} && i = 0, \dots, N-1 \\ & r_{\min} \leq r \leq r_{\max} && i = 0, \dots, N-1 \\ & T_{\text{R,max}} \leq T_{\text{port}}, T_{\text{stbd}} \leq T_{\text{F,max}} && i = 0, \dots, N-1 \\ & \dot{T}_{\text{R,max}} \leq \dot{T}_{\text{port}}, \dot{T}_{\text{stbd}} \leq \dot{T}_{\text{F,max}} && i = 0, \dots, N-1 \\ & \sqrt{(x_O^j - x)^2 + (y_O^j - y)} \geq R_C^j + R_S \end{aligned}$$

where $u_{\min}, u_{\max}, r_{\min}, r_{\max}, \dot{T}_{\text{R,max}}, \dot{T}_{\text{F,max}}$ are constraints for their respective variables, N denotes the number of sampling instants, and $\boldsymbol{Q}, \boldsymbol{R}, \boldsymbol{S}, \boldsymbol{Q}_N$ are weighting matrices. Then, the vector $\boldsymbol{z} = [\boldsymbol{z}_l, \boldsymbol{z}_u, 1]^T$

Algorithm 1 NMPC-based path-following and collision avoidance algorithm.

1: Set number of sampling instants N in the time prediction horizon T_p, and the sampling time T_s.
2: Set weight matrices $\boldsymbol{Q}, \boldsymbol{R}, \boldsymbol{S}, \boldsymbol{Q}_N$
3: Set constraints and slack variable
4: Set state references $\boldsymbol{x}_{i,\text{ref}}$, $\boldsymbol{x}_{N,\text{ref}}$
5: Get the current value $\boldsymbol{x}$
6: Obtain x_O, y_O, and R_O from each obstacle
7: Solve the NLP problem (22), and take the optimal control input vector $[\boldsymbol{u}_0 \dots \boldsymbol{u}_{N-1}]$ and the predicted states $[\boldsymbol{x}_1 \dots \boldsymbol{x}_N]$
8: Grab the element $\boldsymbol{x}_{20}$; the algorithm goes to Step 5.

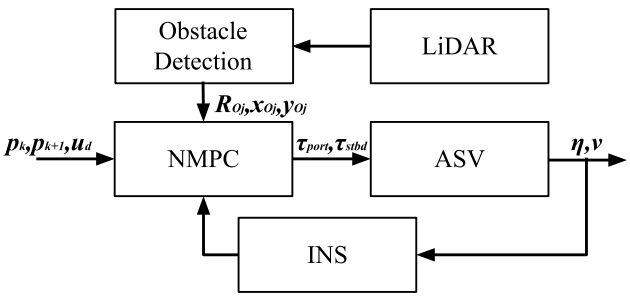

Fig. 2. System architecture of proposed NMPC-based strategy.

contains the lower bound $\boldsymbol{z}_l = [z_{l,1}, z_{l,2}, \dots, z_{l,j}]^T$ and upper bound $\boldsymbol{z}_u = [z_{u,1}, z_{u,2}, \dots, z_{u,j}]^T$ slack variables for each obstacle, where $z_{l,1}, z_{l,2}, \dots, z_{l,j} = R_S$, $z_{u,1}, z_{u,2}, \dots, z_{u,j} = 0$, and $\boldsymbol{S}$ is

$$\boldsymbol{S} = \begin{bmatrix} 0 & 0 & \zeta_l \\ 0 & 0 & \zeta_u \\ \zeta_l^T & \zeta_u^T & 0 \end{bmatrix} \tag{23}$$

with $\zeta_u, \zeta_l \in \mathbb{R}^j$. Moreover, Algorithm 1, contains the NMPC formulation.

4. Practical implementation

In this section, the elements which compose the practical implementation of the proposed NMPC are described. First, the hardware elements of the ASV are presented. Then, the obstacle detection system is introduced. Finally, based on the architecture of the proposed system, shown in Fig. 2, the NMPC algorithm implementation is reported.

4.1. ASV platform

The VTec S-III ASV (see Fig. 3) is selected for the implementation of the NMPC-based path-following and collision avoidance system. The onboard processor is an NVIDIA Jetson TX2 with Robot Operating System (ROS) Kinetic and a Linux environment. A Vectornav VN-300 Inertial Navigation System (INS) is used for position and velocity information, whereas a Velodyne VLP-16 LiDAR provides obstacle detection. The catamaran ASV is propelled by two Blue Robotics T200 thrusters, which are connected to an Arduino Nano. Moreover, the Arduino is connected to the Jetson TX2 via USB for serial communication, and to a remote radio-frequency controller. $R_V = 0.5$ m. Finally, for real-world experiments, the thrust values should be transformed into PWM values by:

$$T_{\text{PWM},i} = \begin{cases} \dfrac{T_{N,i}}{T_{\text{F,max}}} * 385 + 1515, & \text{if } T_{N,i} \geq 0 \\ \dfrac{-T_{N,i}}{T_{\text{R,max}}} * 385 + 1485, & \text{if } T_{N,i} < 0 \end{cases} \tag{24}$$

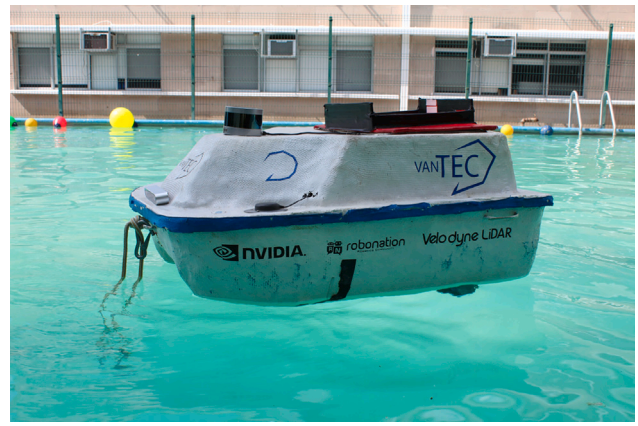

Fig. 3. VTec S-III ASV.

where $T_{N,i}$ is the desired thrust in Newtons of each thruster, $T_{\text{PWM},i}$ is the computed PWM signal of each thruster, and $T_{\text{F,max}} = 36.5$, $T_{\text{R,max}} = -30$ are the maximum forward and reverse values that the thruster can reach, respectively.

4.2. LiDAR-based obstacle detection

The obstacle detection system uses a LiDAR 3D point cloud input, obtained from a Velodyne VLP-16 Puck. Only a close range region of interest, centered at the LiDAR's current location, is considered, thus removing unnecessary background data and accelerating the computation. A crop box filter with the following dimensions in meters was applied: $x_{\min} = -5.0$, $x_{\max} = 5.0$, $y_{\min} = -5.0$, $y_{\max} = 5.0$, $z_{\min} = -0.1$, and $z_{\max} = 0.1$.

Using a grid-based approach similar to Börcs et al. (2017), the remaining point cloud is projected onto the $P_{z=0}$ plane and structured as a 2D grid S, where $s \in S$ represents an individual grid cell of 0.2 m x 0.2 m. The following features are then obtained from each grid cell: $z_{\max}(s)$, $z_{\min}(s)$, center point $\boldsymbol{cp} = [x_L, y_L]^T$ and point density. This information is used to discard individual grids with low density or small relative height ($|z_{\max}(s) - z_{\min}(s)|$).

Objects are formed by clustering adjacent s cells with similar height and density. The data from each s in an object contributes to the overall characteristics and localization of the object. The object center point is the average $\overline{cp}(s)$, and the overall height is determined using the maximum of the $z_{\max}(s)$ and minimum of the $z_{\min}(s)$. To determine the radius of the object, first the $x_{\max}(s)$, $x_{\min}(s)$, $y_{\max}(s)$, and $y_{\min}(s)$ are obtained. Next, the distances from $\overline{cp}(s)$ to each opposing corner of the object are computed using the following euclidean distance formulas

$$c_{bottom} = \sqrt{x_{\min}(s)^2 + y_{\min}(s)^2} \tag{25}$$

$$c_{top} = \sqrt{x_{\max}(s)^2 + y_{\max}(s)^2} \tag{26}$$

The radius of each object j is then obtained by choosing the greatest corner distance

$$R_O^j = \max\{c_{bottom}^j, c_{top}^j\} \tag{27}$$

Finally, each obstacle position is transformed into the NED frame following

$$\begin{bmatrix} x_O^j \\ y_O^j \end{bmatrix} = \begin{bmatrix} \cos(\psi) & -\sin(\psi) \\ \sin(\psi) & \cos(\psi) \end{bmatrix} \begin{bmatrix} x_L^j + l_L \\ y_L^j \end{bmatrix} + \begin{bmatrix} x \\ y \end{bmatrix} \tag{28}$$

where $l_L = 0.25$m is the distance between the LiDAR center and the body frame origin.

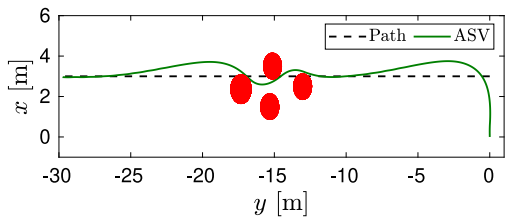

Fig. 4. Simulation I. Reference path, ASV trajectory, and detected obstacles (red). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

4.3. NMPC implementation

Considering the high-performance embedded capability, the acados optimal control framework (Verschueren et al., 2018) is selected to solve the NLP (22). The acados framework has been tested with practical systems such as quadrotors (Carlos et al., 2020) and scale racing cars (Kloeser et al., 2020). Additionally, acados is faster than its predecessor, ACADO (Verschueren et al., 2018). A Real-Time Iteration scheme with Sequential Quadratic Programming and Gauss–Newton Hessian approximation approximately solves the NLP (Carlos et al., 2020). The acados framework outputs embedded C-code which is then implemented as part of a C++ ROS node to be integrated with the ASV software via rostopics.

For experimental evaluation, the NMPC time horizon was set to $T_p = 4$ s, with $N = 400$. Since the LiDAR can run at a frequency of 20 Hz, the frequency of the ROS node was also set to 20 Hz, and the value of $T_{\text{port}}, T_{\text{stbd}}$ was taken at $i = 20$. Furthermore, the safety distance was chosen as $R_S = 0.2$ m, and the constraints were selected as $u_{\min} = -1.5$, $u_{\max} = 1.5$, $r_{\min} = -5.0$, $r_{\max} = 5.0$, $\dot{T}_{\text{R,max}} = -90$, and $\dot{T}_{\text{F,max}} = 90$. The weights were selected experimentally as

$$\boldsymbol{Q} = \text{diag}([0, 2, 2, 0.5, 0, 0, 30, 0, 0.1, 0.001, 0.001]) \tag{29}$$

$$\boldsymbol{R} = \text{diag}([0.001, 0.001]) \tag{30}$$

$$\boldsymbol{\zeta}_u = [0, 0, \dots, 0]^T \in \mathbb{R}^j \tag{31}$$

$$\boldsymbol{\zeta}_l = [50, 50, \dots, 50]^T \in \mathbb{R}^j \tag{32}$$

$$\boldsymbol{Q}_N = \text{diag}([0, 4, 4, 1, 0, 0, 60, 0, 0.2, 0.002, 0.002]) \tag{33}$$

The weights on matrix $\boldsymbol{Q}$ and $\boldsymbol{Q}_N$ were designed to penalize heading in a different direction than the path, the distance away from the path, not maintaining the desired speed, unnecessary/aggressive rotational speeds, and unnecessary/large control effort. Similarly, the weights on $\boldsymbol{R}$ penalize aggressive changes in the control signals, to achieve a smooth thruster signal. Finally, matrices $\boldsymbol{\zeta}_u$ and $\boldsymbol{\zeta}_l$ penalize the closeness to the detected obstacles. Nevertheless, the priority is to avoid obstacles, allowing the ASV to deviate from the reference path and speed if needed.

5. Simulation results

In this section, two simulation runs are described, which allowed to analyze the performance of the proposed strategy in distinct scenarios. Numerical simulations were performed using the dynamic model from Gonzalez-Garcia and Castañeda (2021), and a Linux platform running ROS.

5.1. Simulation I

The first simulation required the robot follow a simple straight path consisting of two waypoints $\boldsymbol{p} = ([3, 0], [3, -30])$. Four obstacles were placed in a diamond formation obstructing the desired path, three of the obstacles had a 0.105 m radius and the remaining one had a

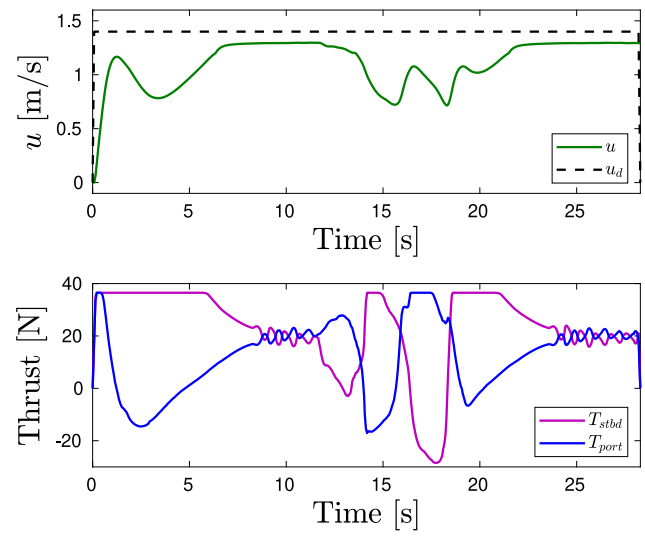

Fig. 5. Simulation I. Speed reference and response (top). Thruster signals (bottom).

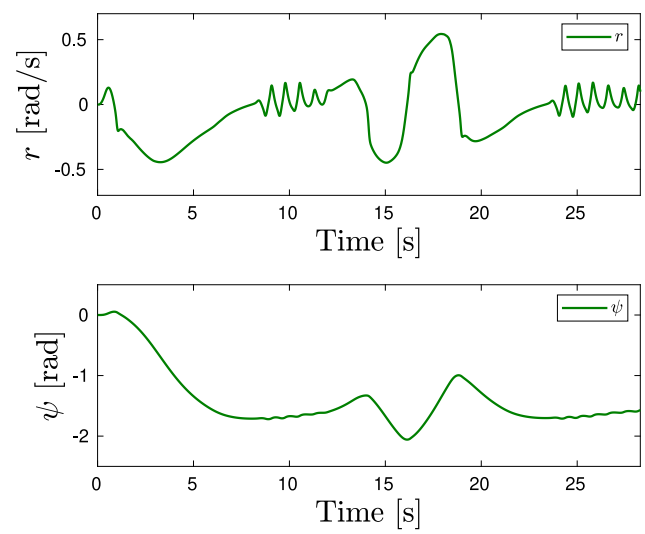

Fig. 6. Simulation I. Yaw rate response (top). Heading angle (bottom).

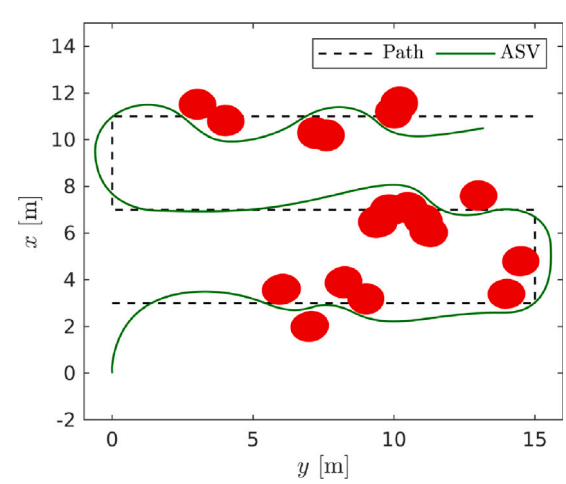

Fig. 7. Simulation II. Reference path, ASV trajectory, and detected obstacles (red). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

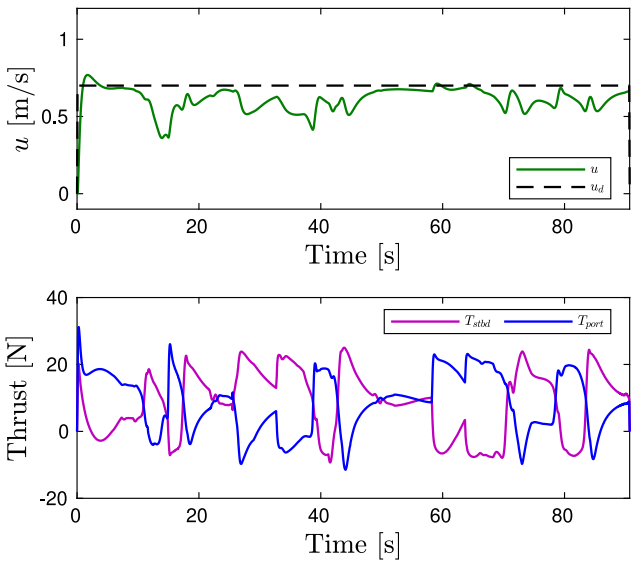

Fig. 8. Simulation II. Speed reference and response (top). Thruster signals (bottom).

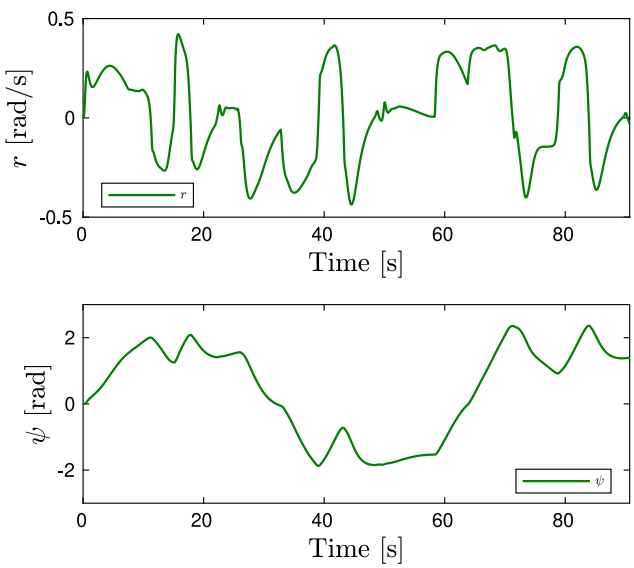

Fig. 9. Simulation II. Yaw rate response (top). Heading angle (bottom).

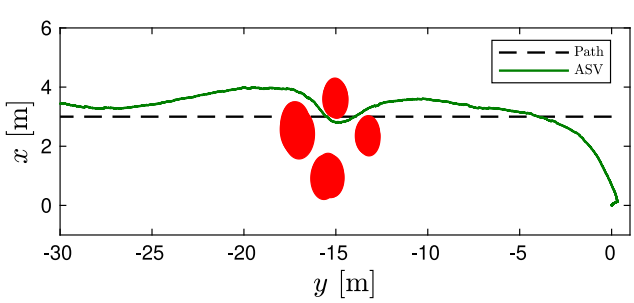

Fig. 10. Experiment. Reference path, ASV trajectory, and detected obstacles (red). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

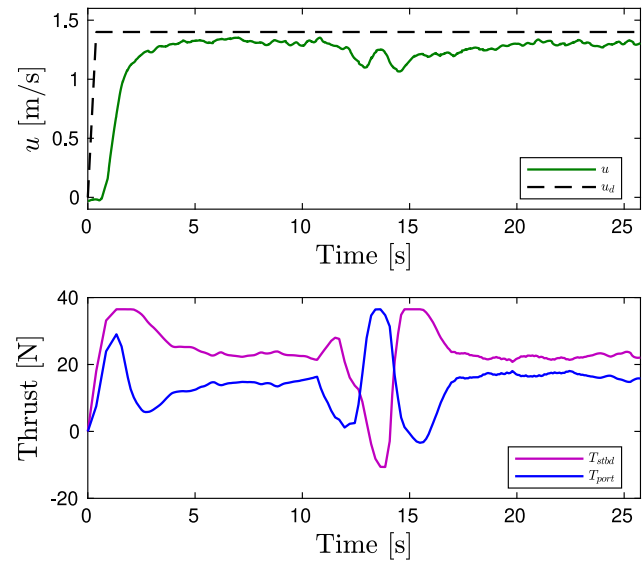

Fig. 11. Experiment. Speed reference and response (top). Thruster signals (bottom).

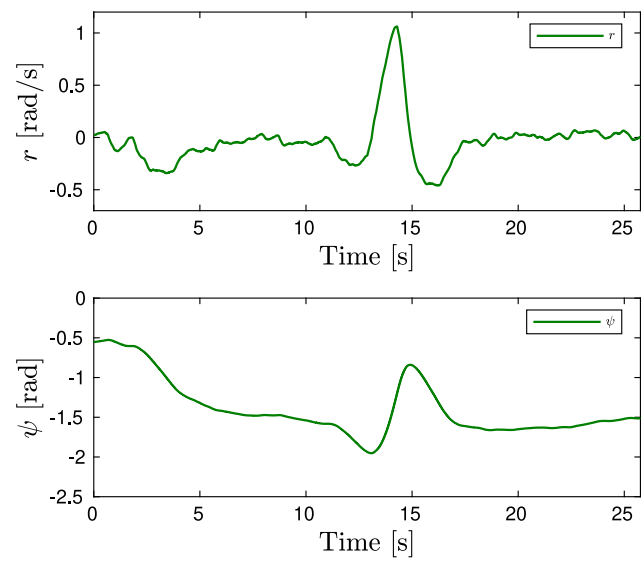

Fig. 12. Experiment. Yaw rate response (top). Heading angle (bottom).

0.195 m radius. The imposed target speed was $u_d = 1.4$, which is close to the maximum speed of the platform. The initial conditions were $x, y, \psi, u, v, r = 0$. The proposed weighting matrices were tuned so that bypassing obstacles is of greater importance than remaining on the predefined path and maintaining desired speed. The simulation results are shown in Fig. 4 . The robot successfully deviated from the path to avoid collision with the obstacles before returning to the desired path. It is evident the turning maneuver was performed to the obstacle side that permitted the vehicle to stay the nearest possible to the desired path. In Fig. 5, the vehicle surge speed and the thrust response are depicted. The surge speed fluctuated while reaching the path and during avoidance actions. Nevertheless, it remained close to the desired speed when following the path. The plot illustrates the proposed capability to reduce the vehicle velocity to facilitate maneuvering. The thruster signals were continuous and the fluctuations correspond to smooth changes in speed and heading while turning. Angular velocity and heading angle, although not directly controlled, are shown in Fig. 6.

a) b) c) d) e)

Fig. 13. Video frames from the experiment. (a) The ASV approaches the obstacles. (b) The ASV avoids the green and right red buoy. (c) The ASV avoids the yellow and right red buoy. (d) The ASV exits the obstacles area. (e) The ASV returns to the path. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

5.2. Simulation II

The second simulation test displays the algorithm in complex scenarios, which are challenging to construct during field experiments. The desired path was composed of waypoints $\boldsymbol{p} = ([3,0],[3,15],[7,15],[7,0],[11,0],[11,15])$, with 18 obstacles of radius 0.105 m located around said path. The desired speed was selected as $u_d = 0.7$, which is a maneuverable speed for the VTec S-III. The initial conditions were once again $x, y, \psi, u, v, r = 0$. When the vehicle was inside a 2 m radius of $\boldsymbol{p}_{k+1}$, the vehicle switched to the next path segment. The path and the ASV trajectory are shown in Fig. 7. The ASV managed to avoid all of the encountered obstacles, while it was capable of smoothly returning to the path segments with no obstacles around them. In Fig. 8, the surge speed response and the thrust are illustrated. The surge speed was kept close to the desired speed, but the relaxed weights allowed it to decrease for short periods to increase manoeuvrability. In addition, the thruster signals were continuous, which is beneficial to achieve a smooth behavior in a practical scenario. Angular velocity and heading angle are illustrated in Fig. 9, depicting the moments where the ASV decided to turn, either to keep following the path or to avoid an obstacle.

6. Experimental results

In this section, to show the effectiveness of the proposed strategy, real-time experiments were conducted. Similar to the first simulation, the boat initial conditions were $x, y, \psi, u, v, r \approx 0$, and it was commanded to follow a path $\boldsymbol{p}_k = [3,0]$ and $\boldsymbol{p}_{k+1} = [3,-30]$, with a constant speed of $u_d = 1.4$ m/s, which is almost the ASV maximum speed. To further demonstrate, the same four buoys in a diamond shape were located such that the path was blocked, requiring avoidance maneuvers. A video of the experimental run can be found here.[1]

In Fig. 10, the reference path, the ASV trajectory, and the obstacles with radius R_C^j are shown. As a remark, the position and radius were detected with the LiDAR at every time-step, between sensor noise and water-induced motion, the detected position and size varied throughout the experiment. Thus, the obstacles in the figure appear to have nonuniform shapes. The vehicle was capable of closely following the path when there were no obstacles, whereas it diverged from the path to successfully evade the buoys. Likewise, the ASV chose to avoid from the side of the obstacle which was closer to the path, hence reducing the distance it had to diverge. A similar feature is seen in Fig. 11, where the speed was kept close to the desired speed during most of the experiment. Nevertheless, the weights were selected to prioritize the safety of the ASV rather than the accuracy of the path-following and speed control. Therefore, the speed was reduced to decrease the maneuvering difficulty, which allowed for safer collision avoidance. Fig. 11 additionally exhibits that the actuator signals were smooth and continuous, which is consequence of the proposed state augmentation. The uncontrolled angular variables, r and ψ, are depicted in Fig. 12. In the same time period the surge speed decreased to facilitate maneuvering ($12 \lessapprox t \lessapprox 17$), the yaw angle and rate deviated, representing that the vehicle turned to avoid collisions. After the evasion maneuver, both variables converged to regions around 0 rad/s and $-\pi/2$, respectively, which allowed the ASV to follow the straight line path. Furthermore, in Fig. 13, snapshots from the experiment illustrate the avoidance maneuvers the ASV performed

This experimental result obtained is similar to the performance of Simulation I. The avoidance procedure followed the same turning directions, leading to similar heading graphs. The NMPC thruster signals were smooth, and the velocity decreased to ease the evasive action. A relevant characteristic of the approach is that the acados implementation allowed to perform the experiment with a frequency of 20 Hz. Likewise, despite model uncertainty and environmental disturbances pertinent to the real-world, the NMPC strategy achieved the path-following and collision avoidance task, along with the speed control.

7. Conclusions

An end-to-end path-following and collision avoidance strategy for an autonomous surface vehicle has been designed. The proposed approach was based on nonlinear model predictive control, capable of following paths and reference velocities, and modifying its trajectory and velocity to elude local static obstacles. The prediction over a finite horizon of the states was obtained by using a nonlinear ASV dynamic model. A cost function has been designed considering the obstacle detection, and restrictions for states and actuators. The objects were detected in real-time with a LiDAR sensor. Moreover, practical and efficient onboard implementation was addressed employing the high-performance embedded optimal control framework, acados. Finally, numerical simulations and field experiments in the presence of multiple buoys demonstrated the performance and advantages of the proposed scheme. Some limitations which could be addressed in future work are to incorporate dynamic obstacles and external disturbances.

CRediT authorship contribution statement

Alejandro Gonzalez-Garcia: Conceptualization, Methodology, Software, Investigation, Writing – original draft. **Ivana Collado-Gonzalez:** Conceptualization, Methodology, Software, Investigation, Writing – original draft. **Rodolfo Cuan-Urquizo:** Methodology, Software. **Carlos Sotelo:** Methodology, Writing – review & editing, Supervision. **David Sotelo:** Methodology, Writing – review & editing, Supervision. **Herman Castañeda:** Methodology, Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

[1] youtu.be/TMijlRej-dg

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.oceaneng.2022.112900.

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